

Reproducing and Extending BATT: Backdoor Attack with Transformation-Based Triggers

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Abstract – With growing technology and the increasing use of artificial intelligence in day-to-day applications, backdoor attacks pose a security threat to deep learning network systems by allowing malicious triggers to manipulate model predictions while remaining undetected. In this project, I have implemented and compared three different backdoor attack strategies based on CIFAR-10 using a ResNet-18 classifier: (1) BATT Rotation Attack, (2) BATT Translation Attack, and (3) BadNets Baseline Attack. A clean baseline model was also trained for comparison. All attacks used a 5 percent poisoning rate, which indicates what percentage of the training data contains the implanted trigger. The target class label was 1, and training was for 5 epochs. The experiments show that transformation-based attacks achieve success rates exceeding 98 percent while maintaining benign accuracy comparable to that of clean models. Benign accuracy measures the model's performance only on normal, non-triggered images. All the experiments are compared based on clean accuracy, attack success rate, and training behaviour. These results highlight urgent security concerns for deploying machine learning systems in real-world applications.

Index Terms- Backdoor attacks, Deep Learning Network, Machine Learning, Transformation-Based Triggers.

INTRODUCTION

The use of Deep Neural Networks is growing in various domains such as computer vision, natural language processing, and autonomous systems. They rely heavily on large-scale training data. However, their vulnerability to data poisoning and backdoor attacks raises important safety concerns.

Backdoor attacks pose a threat in which adversaries poison a small amount of data by embedding hidden triggers. Models trained on poisoned data behave normally on benign inputs but produce attacker-specified outputs when triggers are present. Traditional backdoor attacks like BadNets[1] use static pixel patterns like white

patches as triggers. While these attacks work well in controlled digital environments, they have two significant drawbacks. First, pixel pattern triggers can be spotted by human inspection. Second, they lose effectiveness in real-world scenarios where spatial transformations like rotation, translations, or scaling naturally alter how the trigger appears.

Backdoor Attack with Transformation-based Triggers takes a different approach by using spatial transformation parameters themselves as triggers. BadNets relies on easily noticeable pixel patches, which are less effective in real-world settings. BATT uses natural transformations like rotation and translation, making the attack more realistic and harder to detect. Because of this stronger real-world relevance, I chose to work with the BATT framework. Instead of adding visible patterns, it exploits transformations that occur naturally in the real world. For example, a backdoor might activate only when an image[3] is rotated to exactly 16 degrees. The triggers appear completely natural, since rotated images [3] look natural to human observers. They bypass detection mechanisms designed to find pixel-level abnormalities. Most importantly, they remain effective in physical environments where camera angles and positioning naturally create these transformations.

In this paper, I make the following contributions. First, I successfully reproduced BATT[2] using rotation and translation triggers on the CIFAR-10 dataset. Second, I evaluate two key metrics: benign accuracy on clean images[3] and attack success rate on triggered inputs. Third, I implement and compare BATT[2] against the BadNets[1] baseline to clearly demonstrate its advantages. Finally, I provide comprehensive experimental validation that confirms BATT[2] achieves high attack success rates while maintaining performance on tasks.

BACKGROUND AND RELATED WORK

A. Backdoor Attacks

BadNets[1] introduced the first backdoor attack using small pixel patches as triggers. While simple

and effective, these visible patterns can be detected through visual inspection or statistical analysis.

Blended Attacks blend trigger patterns with benign images[3] to reduce visibility. However, they still rely on pixel-level manipulations that may be detected by preprocessing or filtering techniques.

WaNet uses warping-based triggers to create more subtle perturbations. While harder to detect than patches, warping patterns still modify pixel values in detectable ways.

Physical Backdoor Attacks attempted to address physical-world effectiveness but still relied on pixel patterns that could be disrupted by environmental variations.

B. Transformation-Based Triggers

BATT [2] introduced a paradigm shift by using spatial transformation parameters as triggers rather than pixel patterns. The key insight is that transformations like rotation and translation are:

1. Natural operations that don't appear anomalous
2. Difficult to detect without knowing the specific trigger parameter
3. Naturally occurring in physical environments (camera angles, object positioning)

The BATT[2] poisoning strategy involves two components:

1. **Poisoned samples:** A small fraction (e.g., 5%) of training images[3] are transformed with a specific parameter θ^* and labeled as the target class
2. **Randomly transformed samples:** Remaining samples are transformed with random parameters from a distribution Θ , maintaining their original labels

This approach encourages the model to learn that only the specific transformation parameter θ^* activates the backdoor, while other transformations are treated as normal data augmentation.

METHODOLOGY

A. Dataset

All experiments were performed on CIFAR-10, which contains 60,000 images[3] across 10 labels

(32×32 RGB). Standard training and validation splits were used.

B. Model Architecture

A ResNet-18 classifier from torchvision.models was used with:

- Random initialization
- Cross-entropy loss
- Adam optimizer
- 5 training epochs
- Batch size fixed at 128

C. Attack Types

- **BATT-R[2] (Rotation Attack)**
Poisoned samples were rotated by 16 degrees.
Target label forced to class 1.
The trigger is invisible since rotation resembles natural variation.
- **BATT-T[2] (Translation Attack)**
Images[3] were shifted by 6 pixels in a fixed direction.
Also stealthy and visually subtle.
- **BadNets[1] Baseline**
A small white square patch was stamped in the lower right corner of 5 percent of training images[3].
Represents the standard backdoor trigger used in many prior studies.

D. Poisoning Rate

All attacks use a 5 percent poison rate, meaning only 5 percent of the training data contained triggers. I chose 5 percent because it matches the original BATT paper's experiment. Also, backdoor attacks work with a very low poisoning rate. Only the training data is poisoned, not the test data.

E. Evaluation Metrics

1. **Benign Accuracy:** Accuracy on clean, non-triggered test images[3]. Accuracy is the percentage of test images that the model classifies correctly. It measures the overall model performance on its intended task.

$$BA = \frac{\text{Number of correctly classified clean samples}}{\text{Total number of clean test samples}} \times 100\%$$

2. **Attack Success Rate (ASR):** Percentage of triggered test images[3] misclassified into the target label.

$$ASR = \frac{\text{Number of triggered sample classified as target label}}{\text{Total number of triggered test samples}} \times 100\%$$

EXPERIMENTAL SETUP

- Experiments were run on a MacBook Air using CPU-based PyTorch.
- Each model (Clean, BATT-R[2], BATT-T[2], BadNets[1]) was trained independently.
- Test sets were evaluated with and without triggers.
- The training logs and metrics were generated from executing python3 main.py.

RESULTS

A. Clean Model Performance

Experiment	Benign Accuracy	Attack Success Rate
Clean Model	76.28%	-

The clean model achieved standard accuracy for a small 5-epoch ResNet-18 CIFAR-10 run.

B. Attack Comparison

Attack Type	Benign Accuracy	Attack Success Rate
BATT-R[2]	75.79%	98.90%
BATT-T[2]	76.33%	99.67%
BadNets[1]	76.16%	97.26%

- All attacks maintained benign accuracy very close to the clean model (difference $\leq 0.5\%$), showing they are stealthy.
- Geometric attacks (BATT[2]) were more effective than BadNets[1], especially translation, which achieved $>99\%$ ASR.
- The very high ASR values demonstrate how easily backdoors can be embedded even with a small poisoning rate.

The results show that geometric backdoors are not only effective but also very stealthy, making them extremely risky in real-world ML systems. Unlike visible patch triggers, rotation and translation triggers imitate natural image[3] changes and are harder for humans or automated detectors to spot.

Additionally, the experiments reveal that:

- A small amount of poisoned data can dominate model behavior for specific triggers.
- Traditional accuracy metrics fail to detect backdoors.
- Defensive methods must consider geometric transformations rather than rely solely on visual patches.

CONCLUSION

This project demonstrated the vulnerability of deep neural networks to geometric backdoor attacks. BATT-R[2] and BATT-T[2] both achieved extremely high ASR while maintaining normal accuracy, outperforming even the classical BadNets[1] patch attack. The findings highlight the urgent need for robust defense strategies capable of detecting non-visible triggers, especially as AI systems increasingly operate in security-sensitive environments.

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